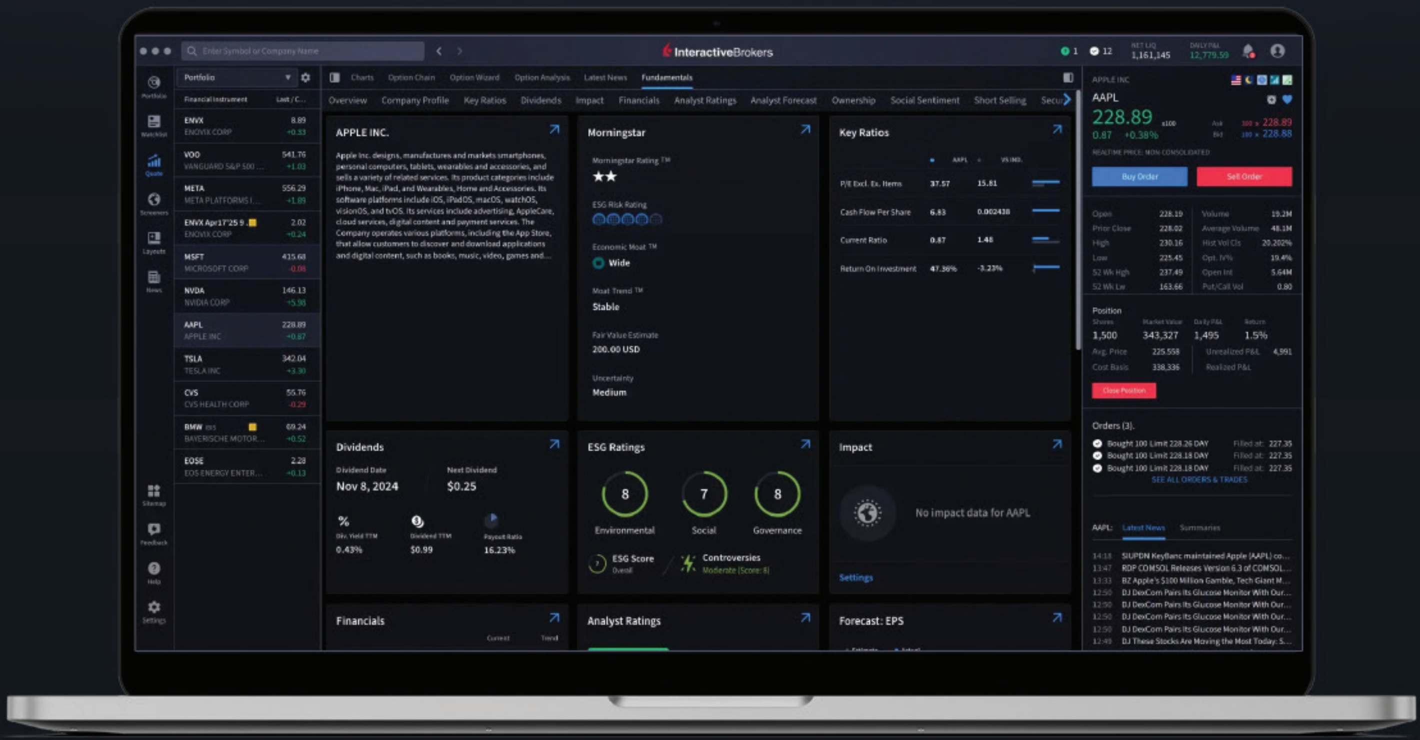
Better Research. Better Decisions.



Take Your Research to the Next Level with Fundamentals Explorer

- Comprehensive global company data
- Easily compare key info to competitors and industry
- Includes key ratios, financials, analyst ratings and forecasts, research reports, and more!

30K+	300+	80+	5.5K+
Companies Covered Worldwide	Data Points per Company	Sources for Newswires and Report	Analyst Ratings from TipRanks

The best-informed investors choose Interactive Brokers

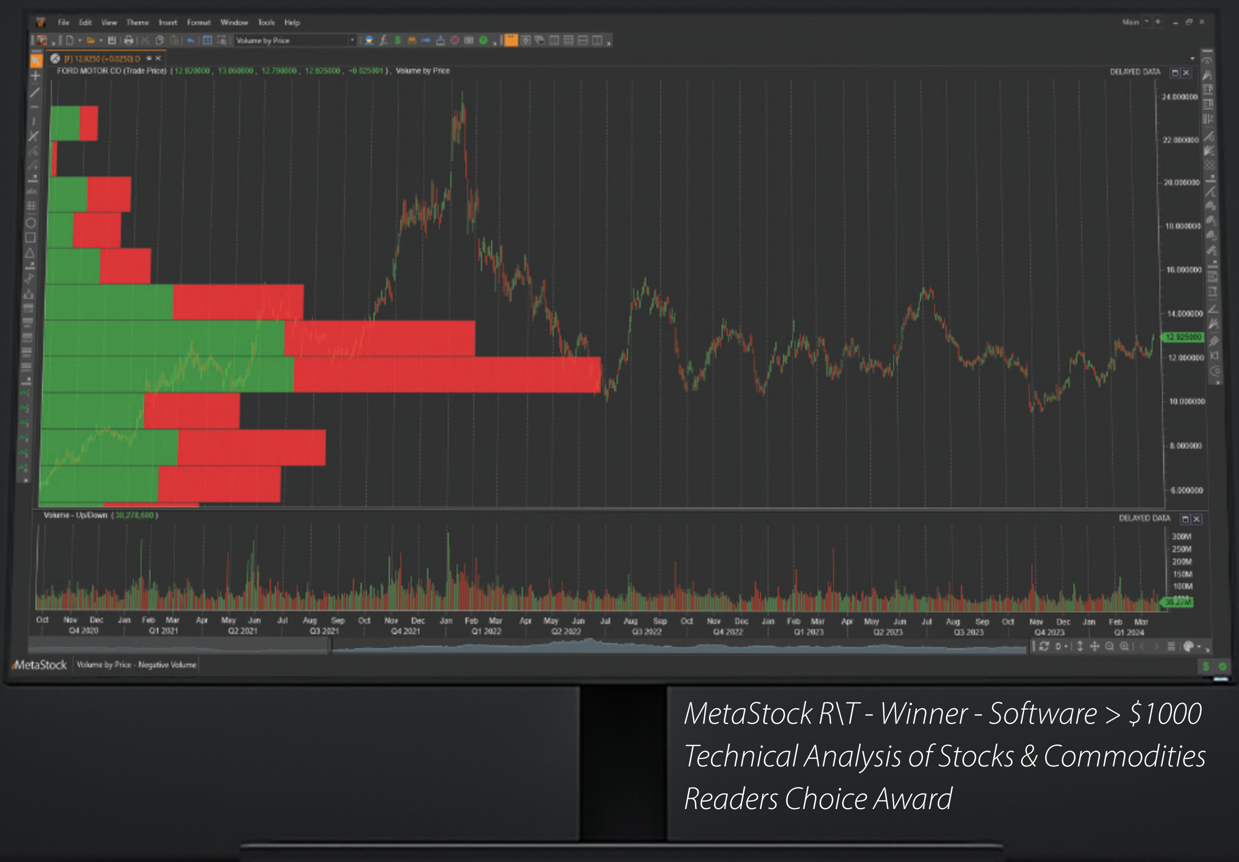
Interactive Brokers

ibkr.com/fund-exp

Member - NYSE, FINRA, SIPC – Any trading symbols displayed are for illustrative purposes only and are not intended to portray recommendations.

01-IB25-1721CH1652

Thank You



MetaStock R\T - Winner - Software > $1000
Technical Analysis of Stocks & Commodities
Readers Choice Award

Again, and again, and again, and again, and again, and ag

Get a free trial at metastock.com/tasc

Copyright © Technical Analysis Inc.

Fundamental Concepts To Elevate Your Trading

Back To Basics

How can you extract meaningful trading information from market data? Here are some crucial insights.

A comment often attributed to Albert Einstein is that if he had an hour to solve a problem, he would spend 55 minutes thinking about the problem and 5 minutes thinking about solutions. Although that may have applied to the physics or scientific realm, the same principle can be applied to trading. That is because market data is a stochastic process, meaning that it is random based on statistics. In fact, the ancient Greek word for stochastic meant "guess," and that often is what we see in trading strategies.

You will not find a single equation in this article. Rather, you will see a basic discussion of some aspects of market data and an overview of how to extract meaningful trading information from this data. Because the concepts are so fundamental, I hope they have a profound affect on your approach to trading.

THE PROBLEM

Market data has a pink spectrum

"Spectrum" means that it contains all frequencies, just like a rainbow. "Pink" means the probability distribution favors the longer wavelengths with respect to amplitude. At the audio level, the sound of rain is pink noise, as opposed to the white noise hiss of an open radio channel. White noise means the amplitude distribution is constant across all wavelengths. Whereas we traders look at bar charts and indicator squiggles in the time domain, "spectrum" implies that the description is in the frequency domain. Fourier theory states that the time descriptions and frequency descriptions are interchangeable.

The Ornstein-Uhlenbeck process describes the random movement of a particle in the presence of friction. It can also describe random price movement where there is knowledge of previous prices. The Ornstein-Uhlenbeck process is similar to the more familiar Black-Scholes model, except that it does not contain a drift rate reflecting expected return. Basically, the Ornstein-Uhlenbeck process says that you can model market prices as a random number generator followed by an exponential moving average (EMA). Try it yourself! You can make a credible-looking price chart by taking an EMA of a random number generator that ranges between 0 and 100. The result will look like prices along a time axis, with a long-term mean of 50. This is the description in the time domain.

The random number generator has a uniform white spectrum. All frequencies are equally likely and have a uniform amplitude. The EMA is a lowpass filter, meaning it will pass low-frequency components and will increasingly attenuate higher and higher frequency components. (I will address filters later in this article.) The rate of attenuation is 6 dB per octave, meaning the amplitude of the spectrum components are halved each time the frequency is doubled. Looking at it another way, the spectrum amplitude components are doubled each time the component wavelength is doubled. That is because wavelength and frequency are reciprocals. We traders prefer to think in terms of wavelengths, like a monthly cycle, for example.

You can easily convince yourself that market data has a pink spectrum. First, construct a chart you like using daily data. Next, create a chart using weekly data. Adjust the horizontal time scale until the second chart looks similar to the first (it's a random process, after all). You will find that the vertical price scale is about five times larger on the second chart than on the first. The two charts will not be exactly the same because we are talking about a stochastic process, and real proof requires lots and lots of sample charts. However, scaling in the price axis in concert with scaling in the time axis is enough to confirm that the

TOPVECTOR/SHUTTERSTOCK

by John F. Ehlers

Copyright © Technical Analysis Inc.

Market data follows a stochastic process and has a pink spectrum.

data has a pink spectrum. This test also demonstrates the relationship between price data and fractals.

So, the pink spectrum explains why the 1932 depression had a bigger swing than prices on my one-minute bars. Of course, this comparison is silly, but it serves to drive home the point that the extent of the pink spectrum is theoretically unbounded. It is far more practical to ask what effect the weekly cycles have on the cycles I am trading using a 15-minute sample rate.

It's all about the cycles, baby

In linear systems theory, it is common to model the data as a pure signal and an additive noise. This enables performance evaluation of the system transfer response with parameters such as signal-to-noise ratios. This model of the data suggests a picture of data as a pure sine wave with additive noise superimposed upon it.

I proved this model of market data to be wrong in my article in the May 2021 issue of Stocks & Commodities, titled "A Technical Description of Market Data for Traders." I showed that if I highpass-filtered the data to eliminate the low-frequency components and establish a zero mean, and then hard-limited the data, I would get about the same waveform after smoothing than I would if I did not hard-limit. Hard-limiting stripped off all the amplitude components in the data. The inescapable conclusion is that the market data basically contains no amplitude noise. The resulting output waveform could be described as being phase modulated or frequency modulated—or just being a composite of a pink spectrum of cycles with no such thing as noise. There are only cycle components in which you are interested for your trading and cycle components that you must eliminate to trade your preferred cycle period effectively.

This description of market data enables the possibility of trading effectively using weekly samples, one-minute samples, or anything in between. The fact that the data follows a stochastic process implies that price patterns or features such as support and resistance or pivot points are entirely subjective and anecdotal.

Aliasing

Market data are sampled data, not a continuous waveform. There must be at least two samples per cycle for analysis of sampled data. The highest possible frequency for sampled data is called the Nyquist frequency. Why, using daily bars, are there higher frequency cycles than allowed by the Nyquist frequency limit? The answer is that the Nyquist frequency is established by the sample rate, not the data.

So, what happens to the cycle energy that is in the data beyond the Nyquist frequency? This higher-frequency data is folded back into the observable spectrum such that the samples fit a much longer wavelength with multiple samples over the aliased wavelength. The pink noise spectrum extends beyond the Nyquist frequency. Thus, the energy at a period just above Nyquist is folded back at nearly full amplitude; what would have been a "one-bar cycle" is folded back to a 4-bar cycle at half amplitude and what would have been a "half-bar cycle" is folded back to an 8-bar cycle at one-quarter amplitude, etc. This is called *aliasing*.

Since market data has a pink spectrum and is sampled data, it is a good rule of thumb not to use a data period less than an eight-bar cycle for analysis due to aliasing distortion.

Fundamentals trump technicals

Every time!

Technicals are governed by statistics because the data follows a stochastic process. On the other hand, fundamentals are identifiable one-time events that tend to skew the statistics. I like to trade the index futures because they average the price variations of individual stocks. Although I trade using technical signals, I have the FOMC calendar on a favorites tab on my computer. Announcements like this are only one example of fundamentals that have a direct impact on prices.

Volume leads price

Really?

This enigma within a fairy tale is based on a macro economic model of the market, and comes complete with a mind-numbing set of rules. This model would hold if the volume, integrated with a smoothing filter, is correlated with price data in the narrow band case. It would also tend to hold if the bandwidth were widened. It doesn't! You can test that yourself.

If one example can prove that volume is not predictive of price, then consider the day session of the emini S&P futures using 15-minute bars. Compare volume to closing prices. Volume has a distinctive 27-bar cycle, with the cycle peaks being at the beginning and of the day. Price has no such easily observable cycle period.

Since volume does not necessarily lead price, it does not have a predictive value. Therefore, the use of volume data probably doesn't help your trading strategy and only serves to contribute to randomization when used in equitick bars or similar nonlinear sampling techniques.

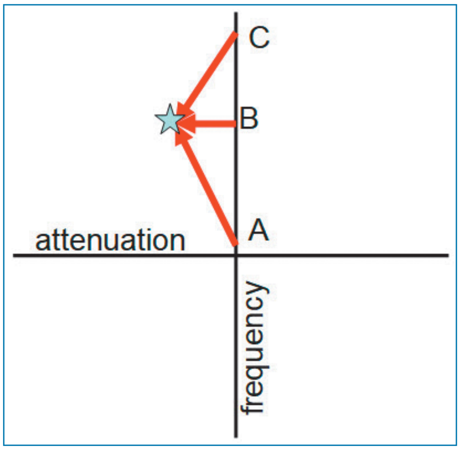


FIGURE 1: PHASE SHIFT. Phase changes nearly 180 degrees per pole over a large-frequency range.

Filters

Since cycles in the pink spectrum are just about all a technical trader has to work with, then filters are about the only tool to be used to extract actionable information. They do this by allowing selected cycle components to pass and to block the undesired cycle components in the spectrum.

It is convenient to think of filters as stonewall filters that entirely pass cycle components on one side of a critical point and completely block cycle components on the other side of the point. Engineers think of filters in terms of frequency. So, a highpass filter passes high-frequency components and rejects low-frequency components. Correspondingly, a lowpass filter passes low-frequency components and rejects high-frequency components. But traders prefer to think in terms of wavelength rather than frequency. *Wavelength is the reciprocal of frequency.* Therefore, a lowpass filter passes the longer wavelengths and rejects the shorter wavelengths. Similarly, a highpass filter passes the shorter wavelengths and rejects the longer wavelengths. A bandpass filter is a combination of a highpass filter and a lowpass filter.

The transfer response of filters is calculated as a ratio of polynomials. If there is only a numerator, the filter is called a *finite impulse response* (FIR) filter. A simple moving average (SMA) filter is one example of a FIR filter. If there is only a denominator in the transfer response, the filter is called an *infinite impulse response* (IIR) filter. Most filters are mixtures of the FIR and IIR types.

An FIR filter operates on a block of data, so the response is referenced to the center of the data block. Therefore, the lag of an FIR filter is about half the block length. All frequency components are lagged the same amount, giving the FIR filter a linear phase response. For example, an eight-bar SMA has a four-bar lag. This is a 22.5 degree phase lag for a 64-bar cycle component, 45-degree lag for a 32-bar cycle component, 90-degree lag for a 16-bar cycle component, etc. This linear phase shift across the data spectrum of the filter output relative to its input means that the input waveshape is not distorted at the output by nonlinear phase dispersion. This is probably the defining feature of a FIR filter. Linear phase delay always results when the filter coefficients are symmetrical about its midpoint. A nonlinear phase transfer results if the coefficients are not symmetrical, as in a weighted moving average.

From the first law of algebra, a polynomial can be factored into zeros of the polynomial. The polynomial only in the denominator of the transfer response is the definition of an IIR filter. An EMA is a first-order IIR filter. There is no divide-by-zero problem however, because the zeros are poles (think of tent poles) in the complex plane. The complex plane has two dimensions to compute filter performance. The horizontal dimension is *gain* or *attenuation*. The vertical plane is *frequency*. The poles can be located anywhere in the plane, but the performance can only be analyzed along the vertical frequency axis using what amounts to vector arithmetic.

IIR filter performance is described with reference to Figure 1. The pole is arbitrarily placed at an intersection of the selected attenuation and frequency. When the input data frequency is at a low-frequency point A, the vector length is long and the phase angle is nearly +90 degrees. Because the vector is long and is in the denominator, the transfer response is small, meaning the filter attenuation is large. When the input data frequency is at point B, the phase angle is zero and the vector length has its minimum value. When the input data frequency is at a high-frequency point C, the phase angle is near –90 degrees and its length is again long. The positions of the various poles of an IIR filter determine its filter shape. Phase shift across the frequency response of a filter is unavoidable.

Real filters are not stonewall filters. Their attenuation rate is 6 dB per octave per pole. So, a second-order IIR filter has an attenuation rate of 12 dB per octave, and so on. The filter output phase shift is nearly 180 degrees across the entire data spectrum for each filter pole. Therefore, when the order of the IIR polynomial is increased, the rate of phase shift increases accordingly. This increased phase shift is why the use of higher-order filters is not advised.

Fundamentals are identifiable one-time events that tend to skew the statistics.

After all, a 180-degree phase flip gives you exactly the wrong answer.

Conventional indicators

Indicators are just filters, often with special scaling features. For example, an MACD is basically a bandpass filter comprised of EMA lowpass filters and a highpass filter consisting of the difference of the EMAs. The EMA difference is a first-order highpass filter, so its attenuation rate is 6 dB per octave as the wavelength is increased. Remembering that the data spectrum has a 6 dB per octave gain as the wavelength is increased, the net effect is only that the data spectrum has been flattened to a white spectrum. The long wavelength data has not been filtered except at zero frequency. This means that the MACD is not helpful in isolating the preferred cycle band for trading.

The RSI indicator is computed as the sum of closes up less the sum of closes down. This difference gives it the characteristic of a first-order highpass filter and the summations serves as the lowpass filter function. The RSI also includes scaling that varies the probability distribution of the filtered output. Since the RSI has only a first-order highpass filter, it also is not helpful in isolating the preferred cycle band for trading.

The *stochastics indicator* is computed as the difference between the last data point and the lowest data point within the observation window. This difference gives it the characteristic of a first-order highpass filter and averaging serves as the lowpass filter function. (Note the term "stochastics" here refers to the name of the popular indicator, not to the generic definition of the word meaning "random" as discussed earlier.) The stochastics indicator also includes scaling that alters the probability distribution of the filtered output. Since the stochastics indicator has only a first-order highpass filter, it also is not helpful in isolating the preferred cycle band for trading.

The solution

Market data follows a stochastic process and has a pink spectrum. There are no special identifying features other than the spectrum consists of cycle components. We can select a relatively narrow band of cycles that fit our trading style or according to the preferred amount of trading activity. We can select that band of frequencies with a bandpass filter, most likely comprised of a highpass filter in tandem with a lowpass filter.

The highpass filter must be at least a second-order filter to filter out the undesired long wave cycle components of the pink spectrum. But to minimize phase shift at the filter output, the highpass filter should not be higher than a second-order filter. The lowpass filter can be a Hann windowed FIR filter or a *projected moving average* (PMA) if lag is to be minimized. Alternatively, the lowpass filter could be a SuperSmoother IIR filter or an UltimateSmoother IIR filter to minimize lag. Filter bandwidth should be on the order of one octave to provide selectivity while keeping a low rate of phase shift across the band.

Because the concepts are so fundamental, I hope they have a profound affect on your approach to trading.

The only information to work with in the bandpass of data is the cycle amplitude and phase. The rate-of-change crossings of zero identify the peaks and valleys of the cycle, which are the ideal entries and exits for swing, or reversion to the mean, trading. Since RMS and standard deviation are synonymous with waveforms having a zero mean, the cycle divided by its RMS value can be displayed as scaled in standard deviations.

If trading to capture the expected return is preferred, follow a Black-Scholes model. There is a plethora of option strategies to do this.

Or: You could trust to luck and trade the trend.

John Ehlers is a retired electrical engineer and a retired technical analyst, specializing in the application of DSP (digital signal processing) to trading. For more information, see www.mesasoftware.com.

Further reading

Ehlers, John [2021]. "A Technical Description Of Market Data For Traders," *Technical Analysis of* Stocks & Commodities, Volume 39: May.

_____ [2025]. "Removing The Lag In Moving Averages," *Technical Analysis of* Stocks & Commodities, Volume 43: March.

_____ [2014]. "Predictive And Successful Indicators," *Technical Analysis of* Stocks & Commodities, Volume 32: January.

_____ [2024]. "The Ultimate Smoother," *Technical Analysis of* Stocks & Commodities, Volume 42: April.

_____ [2004]. *Cybernetic Analysis For Stocks And Futures*, John Wiley & Sons.

_____ [2013]. *Cycle Analytics For Traders*, John Wiley & Sons.

Find similar articles online at Traders.com

NINJATRADER

TRADE FUTURES WITH A LEADING BROKER

Explore the futures markets with our award-winning platform with $50 intraday margins.

UNCOVER YOUR OPPORTUNITY
in futures trading with NinjaTrader:

- Simple pricing
- Low margins & commissions
- Risk-free simulated trading
- 1,000+ third-party tools & apps
- Daily livestreams & on-demand videos
- 100+ technical analysis tools
- 24/5 live support
- No deposit minimum requirement

BETTER FUTURES START HERE.
Open your free account today.

Futures trading contains substantial risk and is not suitable for every investor.

NinjaTrader.com/TASC

TIMELY ABLETREND 8.0 SWEET SPOT
STEP-BY-STEP SIGNAL POSTINGS AS THEY APPEAR

Free AbleTrend Fresh Signals for META

Timely Exit Signals for AbleTrend Users

AbleTrend 30-Day RISK-FREE Trial
Act Now - Limited Time Offer!

- Sign up now to receive FREE real-time email alerts for fresh AbleTrend signals as they appear.
- Stay informed about uptrend stocks at their dips and view these signals in progress.

DON'T MISS THIS OPPORTUNITY TO TAKE CONTROL OF YOUR FINANCIAL FUTURE.
ACT NOW BEFORE THIS LIMITED TIME OFFER EXPIRES!

READERS' CHOICE AWARD

UNLOCK INCREDIBLE BENEFITS WHEN YOU SIGN UP TODAY:

- **70% OFF AbleTrend 8.0** one-month trial (just $39, regularly $130)
- **1 FREE Year of Access** to AbleTrend's Best Stocks to Buy NOW
- **FREE TIMELY EXIT Signals** - take the guesswork out of selling
- **FREE Buy-on-Blue/Sell-on-Red Signals** - our legendary system
- **FREE Market Readiness Insights** - see if the market is ready to launch, near launch, or signaling a downtrend warning
- **FREE One-on-One Consultation** - Gain lifetime financial management skills with expert guidance at no extra cost
- **FREE Members-Only Webinars** - Continuing education every Thursday at 5:00 PM ET
- **FREE Technical Support** - Exceptional, friendly assistance from our acclaimed support team

ABLESYS SINCE 1994

Trustpilot

SPECIAL SALE ONLY, RISK-FREE MONEY-BACK GUARANTEE

YOUR SATISFACTION IS 100% GUARANTEED!

www.ablesys.com/SCM

TRY ABLETREND WITH 70% OFF
Use promo code: SCNY25

©2025 Ablesys Corp. • 8407 Central Ave, Ste 1882, Newark, CA, 94560

sales@ablesys.com

THESE RESULTS ARE BASED ON SIMULATED OR HYPOTHETICAL PERFORMANCE RESULTS THAT HAVE CERTAIN INHERENT LIMITATIONS. UNLIKE THE RESULTS SHOWN IN AN ACTUAL PERFORMANCE RECORD, THESE RESULTS DO NOT REPRESENT ACTUAL TRADING. ALSO, BECAUSE THESE TRADES HAVE NOT ACTUALLY BEEN EXECUTED, THESE RESULTS MAY HAVE UNDER-OR OVER-COMPENSATED FOR THE IMPACT, IF ANY, OF CERTAIN MARKET FACTORS, SUCH AS LACK OF LIQUIDITY. SIMULATED OR HYPOTHETICAL TRADING PROGRAMS IN GENERAL ARE ALSO SUBJECT TO THE FACT THAT THEY ARE DESIGNED WITH THE BENEFIT OF HINDSIGHT. NO REPRESENTATION IS BEING MADE THAT ANY ACCOUNT WILL OR IS LIKELY TO ACHIEVE PROFITS OR LOSSES SIMILAR TO THESE BEING SHOWN. THE TESTIMONIAL MAY NOT BE REPRESENTATIVE OF THE EXPERIENCE OF OTHER CLIENTS AND THE TESTIMONIAL IS NO GUARANTEE OF FUTURE PERFORMANCE OR SUCCESS. TECHNICAL ANALYSIS OF STOCKS & COMMODITIES LOGO AND AWARD ARE TRADEMARKS OF TECHNICAL ANALYSIS, INC.

SUBSCRIBE OR RENEW TODAY!

Every Stocks & Commodities subscription (regular and digital) includes:

- **Full access to our Digital Edition**
  The complete magazine as a PDF you can download.
- **Full access to our Digital Archives**
  That's 35 years' worth of content!
- **Complete access to WorkingMoney.com**
  The information you need to invest smartly and successfully.
- **Access to Traders.com Advantage**
  Insights, tips and techniques that can help you trade smarter.

1 year $89.99

2 years........... $149.99

3 years........... $199.99

PROFESSIONAL TRADERS' STARTER KIT

A 5-year subscription to S&C magazine that includes everything above PLUS a free* book, Charting The Stock Market: The Wyckoff Method, all for a price that saves you $150 off the year-by-year price! *Shipping & handling charges apply for foreign orders.

5 years......... $299.99

That's around $5 a month!

Visit **www.Traders.com** to find out more!

Email: Circ@Traders.com • Phone: 206-938-0570 facebook.com/STOCKSandCOMMODITIES @STOCKSandCOMM